Hubbard Mean Field

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1 Mean Field Hamiltonian

We begin with the Hubabrd term

$$H = U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i,\downarrow} \tag{1}$$

with $\hat{n}_{i\sigma} = c^\dagger_{i\sigma} c_{i\sigma}$. We then treat is in the mean field approach and write

$$\hat{n}_{i\sigma} = \langle n_{i\sigma} \rangle + \delta n_{i\sigma} \tag{2}$$

we plug in (dropping second order fluctuation) to get

$$\begin{aligned} H &= U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \\ &= U \sum_i \left(\langle n_{i\uparrow} \rangle + \delta n_{i\uparrow} \right) \left(\langle n_{i\downarrow} \rangle + \delta n_{i\downarrow} \right) \\ &= U \sum_i \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle + \langle n_{i\uparrow} \rangle \delta n_{i\downarrow} + \delta n_{i\uparrow} \langle n_{i\downarrow} \rangle \end{aligned} \tag{3}$$

Lets replace $\delta n_{i\sigma} = \hat{n}_{i\sigma} - \langle n_{i\sigma} \rangle$

$$\begin{aligned} H &= U \sum_i \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle + \langle n_{i\uparrow} \rangle (\hat{n}_{i\downarrow} - \langle n_{i\downarrow} \rangle) + (\hat{n}_{i\uparrow} - \langle n_{i\uparrow} \rangle) \langle n_{i\downarrow} \rangle \\ &= U \sum_i \langle n_{i\uparrow} \rangle \hat{n}_{i\downarrow} + \langle n_{i\downarrow} \rangle \hat{n}_{i\uparrow} - \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle \end{aligned} \tag{4}$$

Lets introduce $m_i = \langle n_{i\uparrow} \rangle - \langle n_{i\downarrow} \rangle$ and $\langle n_i \rangle = \langle n_{i\uparrow} \rangle + \langle n_{i\downarrow} \rangle$ and rewrite our Hamiltonian to depend on these terms. We can write

$$\langle n_{i\uparrow} \rangle = \frac{\langle n_i \rangle + m_i}{2} \quad \langle n_{i\downarrow} \rangle = \frac{\langle n_i \rangle - m_i}{2} \tag{5}$$

we plug in to get

$$H = \frac{U}{2} \sum_i m_i \left[\hat{n}_{i\downarrow} - \hat{n}_{i\uparrow} \right] + \langle n_i \rangle \left[\hat{n}_{i\downarrow} + \hat{n}_{i\uparrow} \right] + C \tag{6}$$

Where $C = -\langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle$. The second term depends on the occupancy which we fix to be 10/12 for double layer.

$$H = \frac{U}{2} \sum_i m_i \left[\hat{n}_{i\downarrow} - \hat{n}_{i\uparrow} \right] \tag{7}$$

In the z direction we can write $\psi^\dagger \sigma_z \psi = \hat{n}_{i\downarrow} - \hat{n}_{i\uparrow}$ to write

$$H = -\frac{U}{2} \sum_i m_i \hat{\psi}^\dagger \hat{\sigma}_z \hat{\psi} = -\frac{U}{2} \sum_i \hat{\psi}^\dagger m_i \hat{\sigma}_z \hat{\psi} \tag{8}$$

So far we have assumed magnetization is along z. For an arbitrary direction we replace

$$H = -\frac{U}{2} \sum_i \hat{\psi}_i^\dagger \boldsymbol{m}_i \cdot \hat{\boldsymbol{\sigma}} \hat{\psi}_i \tag{9}$$

In our code its easier to write S $= \frac{m}{2} \cdot \hat{n}$. Since $m = Tr(\rho\sigma)$ this includes all orbital contributions. Lastly we can assume translational invariance and drop the sum over lattice points.

$$H = -US\psi^\dagger \hat{n} \cdot \hat{\sigma} \psi \tag{10}$$

Where ψ is the t2g orbital. Since we calculate the MCA by $E(\hat{n}) - E(\hat{z})$ we need to include the constant

$$\begin{aligned} \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle &= (\langle n_i \rangle /2 + S)(\langle n_i \rangle /2 - S) \qquad (11) \\ &= \left(\frac{\langle n_i \rangle}{2} \right)^2 - S^2 \end{aligned}$$

again the filling term is constant and will not effect MCA. Our complete Hamiltonian is then

$$H = -US\psi^\dagger \hat{n} \cdot \hat{\sigma} \psi + US^2 \tag{12}$$

Written in explicit matrix form (spin major) we can write

$$H = -US \left[\hat{n} \cdot \hat{\sigma} \otimes I_3 \right] + US^2 I_6 \tag{13}$$

This constant shift can be included as just an addition to the total energy after diagonalization. So we can solve

$$H' = -US \left[\hat{n} \cdot \hat{\sigma} \otimes I_3 \right] \tag{14}$$

With

$$E(\hat{n}) = E'(\hat{n}) + US^2 \tag{15}$$

2 Calculating the magnetization

In the mean field approach we replace the magnetization with the average. We calculate averages in the grand canonical ensemble

$$m = \mathrm{Tr}(\hat{m}\hat{\rho}) \tag{16}$$

where the density matrix for the grand canonical ensemble is

$$\hat{\rho} = \frac{e^{-\beta(\hat{H} - \mu \hat{N})}}{Z} \tag{17}$$

with $Z = \mathrm{Tr}\ e^{-\beta(H - \mu N)}$. The Hamiltonian is non-interacting so it can be written on the energy basis

$$\hat{H} = \sum_n \varepsilon_n d_n^\dagger d_n \tag{18}$$

where the creation operator in this basis is defined by $d_n^\dagger = \sum_k \langle k|n \rangle c_k^\dagger$. Our fock space is defined by the occupation number in each energy state, for our fermion system this is either zero or one. We label the Fock state like $|\{n_k\}\rangle$. These are eigenstates of the Hamiltonian

$$\hat{H}|\{n_k\}\rangle = \sum_j \varepsilon_j \hat{d}_j^\dagger \hat{d}_j \left(\prod_k \left(d_k^\dagger \right)^{n_k} \right) \tag{19}$$

For $j \neq k$ the commutation is zero and we can flip the two to normal order which results in zero. The only non zero is for when $j = k$. Therefore for a single fock state we can write

$$\hat{H}|\{n_k\}\rangle = \sum_j \varepsilon_j n_j |\{n_k\}\rangle \tag{20}$$

the same reasoning allows us to write

$$\hat{N}|\{n_k\}\rangle = \sum_j n_j |\{n_k\}\rangle \tag{21}$$

using these results we find the partition function

$$\begin{aligned} Z &= \sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\langle\{n_k\}|e^{-\beta(\hat{H}-\mu\hat{N})}|\{n_k\}\rangle \\ &= \sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\langle\{n_k\}|e^{-\sum_j\beta(\varepsilon_j-\mu)n_j}|\{n_k\}\rangle \\ &= \sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\prod_j\langle\{n_k\}|e^{-\beta(\varepsilon_j-\mu)n_j}|\{n_k\}\rangle \end{aligned} \tag{22}$$

The sums out front go from 0, 1 since these are fermions. This means that the sums decouple and we can bring the product outside. Then we evaluate one sum from 0 to 1 and we get

$$Z = \prod_j \left(1 + e^{-\beta(\varepsilon_j-\mu)}\right) \tag{23}$$

Next we are interested in the density

$$\begin{aligned} \langle d_m^\dagger d_n\rangle &= \frac{\mathrm{Tr}\left(e^{-\beta(\hat{H}-\mu\hat{N})}\hat{d}_m^\dagger d_n\right)}{Z} \\ &= \frac{1}{Z}\sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\langle\{n_k\}|e^{-\beta(\hat{H}-\mu\hat{N})}\hat{d}_m^\dagger d_n|\{n_k\}\rangle \end{aligned} \tag{24}$$

Since our $\hat{H}$ and $\hat{N}$ are diagonal the density matrix is diagonal in this basis and $m = n$.

$$\begin{aligned} \langle d_n^\dagger d_n\rangle &= \frac{1}{Z}\sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\langle\{n_k\}|e^{-\beta(\hat{H}-\mu\hat{N})}\hat{d}_n^\dagger d_n|\{n_k\}\rangle \\ &= \frac{1}{Z}\sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\langle\{n_k\}|e^{-\beta(\hat{H}-\mu\hat{N})}n_n|\{n_k\}\rangle \\ &= \frac{1}{Z}\sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\langle\{n_k\}|e^{-\sum_j\beta(\varepsilon_j-\mu)n_j}n_n|\{n_k\}\rangle \\ &= \frac{1}{Z}\sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\langle\{n_k\}|n_n\prod_j e^{-\beta(\varepsilon_j-\mu)n_j}|\{n_k\}\rangle \end{aligned} \tag{25}$$

We can cancel out all $j \neq n$ terms from the partition function in the denominator. Hence we will get

$$\begin{aligned} \langle d_n^\dagger d_n\rangle &= \frac{\sum_{n=0}^1 n_n e^{-\beta(\varepsilon_n-\mu)n_n}}{1+e^{-\beta(\varepsilon_n-\mu)}} \\ &= \frac{e^{-\beta(\varepsilon_n-\mu)}}{1+e^{-\beta(\varepsilon_n-\mu)}} \\ &= f(\varepsilon_n-\mu) \end{aligned} \tag{26}$$

where f is the fermi dirac distribution. Now we perform a change of basis to the t2g basis. Recall $d_n^\dagger = \sum_k \langle k|n\rangle c_k^\dagger$. From the definition of the density matrix we can write

$$\begin{aligned}\rho_{ab} = \langle c_a^\dagger c_b\rangle &= \langle \left(\sum_k \langle a|k\rangle d_k^\dagger\right)\left(\sum_q \langle q|b\rangle d_q\right)\rangle \\ &= \sum_{kq} \langle a|k\rangle\langle q|b\rangle\langle d_k^\dagger d_q\rangle \\ &= \sum_{kq} \langle a|l\rangle\langle q|b\rangle f(\varepsilon_k - \mu)\delta_{kq} \\ &= \sum_n \langle a|n\rangle\langle n|b\rangle f(\varepsilon_k - \mu)\end{aligned} \tag{27}$$

Where $|n\rangle$ labels the eigenstates and a, b labels the t2g,spin band.

2.1 Magnetization

Now we evaluate the magnetization

$$\begin{aligned}\boldsymbol{m} = \mathrm{Tr}(\rho\boldsymbol{\sigma}) &= \sum_n \langle n|\rho\boldsymbol{\sigma}|n\rangle \\ &= \sum_{n,b} \langle n|\rho|b\rangle\langle b|\boldsymbol{\sigma}|n\rangle \\ &= \sum_{n,b,a,\alpha} \langle n|a\rangle\rho_{ab}\sigma_{b\alpha}\langle\alpha|n\rangle \\ &= \sum_{n,b,a,\alpha} \langle\alpha|n\rangle\langle n|a\rangle\rho_{ab}\sigma_{b\alpha} \\ &= \sum_{b,a,\alpha} \langle\alpha|a\rangle\rho_{ab}\sigma_{b\alpha} \\ &= \sum_{ba} \rho_{ab}\sigma_{ba} \\ &= \mathrm{Tr}(\rho\cdot\sigma)\end{aligned} \tag{28}$$

This allows us to write the components of m as

$$m_i = \sum_{ba} \rho_{ab}\sigma_{ba}^i = \sum_{ba}\sum_n f(\varepsilon_n - \mu)\langle a|n\rangle\langle n|b\rangle\sigma_{ba}^i \tag{29}$$

Where again n are the energy eigenstates and a, b are the t2g,spin basis. When we diagonalize we get eigenvectors like

$$|n\rangle = \sum_a \langle a|n\rangle|a\rangle = \sum_a c_a|a\rangle \tag{30}$$

Hence we can rewrite the magnetization as

$$m_i = \sum_{ba}\sum_n f(\varepsilon_n - \mu)c_a c_b^* \sigma_{ba}^i \tag{31}$$

To simplify each of the cases further we can plug in the corresponding pauli matrix.

$$m_x = \sum_n\sum_{ab} f(\varepsilon_n - \mu)c_a c_b^*\sigma_{ba}^x = \sum_n f(\varepsilon_n - \mu)(c_1 c_2^* + c_2 c_1^*) \tag{32}$$

Where we can use the fact that with $z = a + bi$ and $y = c + di$ and $z^*y = (a - bi)(c + di) = ac + bd + adi - bci$

$$\begin{aligned} zy^* + z^*y &= (a + bi)(c - di) + (a - bi)(c + di) \\ &= ac - adi + bci + bd + ac + adi - bci + bd \\ &= 2(ac + bd) = 2\mathrm{Re}(z^*y) \end{aligned} \tag{33}$$

hence with $1 = \uparrow$ and $2 = \downarrow$ we can write

$$m_x = 2\sum_n f(\varepsilon_n - \mu)\mathrm{Re}(c_\uparrow^* c_\downarrow) \tag{34}$$

Likewise for m_y we can write

$$m_y = \sum_n f(\varepsilon_n - \mu)(c_1 c_2^* - c_2 c_1^*)i \tag{35}$$

Where

$$\begin{aligned} -(yz^* - zy^*) &= -((c + di)(a - bi) - (a + bi)(c - di)) \\ &= -(ca - bci + dai + db - (ac - adi + bci + bd)) \\ &= -2(ad - bc)i = -2\mathrm{Im}(z^*y)i \end{aligned} \tag{36}$$

Likewise we can write

$$m_y = 2\sum_n f(\varepsilon_n - \mu)\mathrm{Im}(c_\uparrow^* c_\downarrow) \tag{37}$$

Lastly for m_z we can write

$$m_z = \sum_n f(\varepsilon_n - \mu)(|c_\uparrow|^2 - |c_\downarrow|^2) \tag{38}$$

Now the projection is

$$m_{\hat{n}} = \boldsymbol{m} \cdot \hat{n} = m_x n_x + m_y n_y + m_z n_z \tag{39}$$

3 Total Energy

The total energy is given by

$$\begin{aligned} E &= \mathrm{Tr}(\hat{H}\hat{\rho}) && (40) \\ &= \sum_n \langle n|H\rho|n\rangle && (41) \\ &= \sum_{n,m} \langle n|H|m\rangle\langle m|\rho|n\rangle \\ &= \sum_{n,m} \varepsilon_n \delta_{nm} f(\varepsilon_n - \mu) \\ &= \sum_n \varepsilon_n f(\varepsilon_n - \mu) \end{aligned}$$

4 Number of particles and chemical potential

4.1 Number of particles

The number of particles per unit cell is given by

$$N = \frac{1}{N_k} \sum_{n,k} f(\varepsilon_{n,k} - \mu) = 10 \tag{42}$$

Where we have 5/6 electrons in the t2g. For two layers each unit cell should have 10 electrons. We can employ Brent's method of root finding to find the chemical potential by solving this equation.

$$N(\mu) - 10 = 0 \tag{43}$$

5 Kanamori Hamiltonian

To capture the complete physics we extend out model to the Kanamori Hamiltonian

$$\begin{aligned} H_K = U \sum_m \hat{n}_{m\uparrow}\hat{n}_{m\downarrow} + U' \sum_{m\neq m'} \hat{n}_{m\uparrow}\hat{n}_{m'\downarrow} + (U' - J) \sum_{m<m',\sigma} \hat{n}_{m\sigma}\hat{n}_{m'\sigma} \\ - J \sum_{m\neq m'} d^{\dagger}_{m\uparrow} d_{m\downarrow} d^{\dagger}_{m'\downarrow} d_{m'\uparrow} + J \sum_{m\neq m'} d^{\dagger}_{m\uparrow} d^{\dagger}_{m\downarrow} d_{m'\downarrow} d_{m'\uparrow} \end{aligned} \tag{44}$$

We have already looked at the MF treatment of the first term above.

5.1 Onsite Repulsion

$$H = U \sum_m \langle n_{m\uparrow}\rangle \hat{n}_{m\downarrow} + \langle n_{m\downarrow}\rangle \hat{n}_{m\uparrow} - \langle n_{m\uparrow}\rangle\langle n_{m\downarrow}\rangle \tag{45}$$

5.2 Interorbital Repulsion

5.2.1 Opposite Spin

$$H = U' \sum_{m\neq m'} \hat{n}_{m\uparrow}\hat{n}_{m'\downarrow} \tag{46}$$

As before each operator is written as

$$\hat{n}_{m\sigma} = \langle n_{m\sigma}\rangle + \delta n_{m\sigma} \tag{47}$$

which allows us to write

$$\begin{aligned} H &= U' \sum_{m\neq m'} [\langle n_{m\uparrow}\rangle + \delta n_{m\uparrow}][\langle n_{m'\downarrow}\rangle + \delta n_{m'\downarrow}] \\ &= U' \sum_{m\neq m'} \langle n_{m\uparrow}\rangle\langle n_{m'\downarrow}\rangle + \langle n_{m\uparrow}\rangle \delta n_{m'\downarrow} + \langle n_{m'\downarrow}\rangle \delta n_{m\uparrow} \end{aligned} \tag{48}$$

with $\delta n_{m\sigma} = \hat{n}_{m\sigma} - \langle n_{m\sigma}\rangle$

$$\begin{aligned} H &= U' \sum_{m\neq m'} \langle n_{m\uparrow}\rangle\langle n_{m'\downarrow}\rangle + \langle n_{m\uparrow}\rangle[\hat{n}_{m'\downarrow} - \langle n_{m'\downarrow}\rangle] + \langle n_{m'\downarrow}\rangle[\hat{n}_{m\uparrow} - \langle n_{m\uparrow}\rangle] \\ &= U' \sum_{m\neq m'} \langle n_{m\uparrow}\rangle \hat{n}_{m'\downarrow} + \langle n_{m'\downarrow}\rangle \hat{n}_{m\uparrow} - \langle n_{m\uparrow}\rangle\langle n_{m'\downarrow}\rangle \end{aligned} \tag{49}$$

5.2.2 Same Spin

The next term is

$$
\begin{aligned}
H &= (U' - J) \sum_{m<m',\sigma} \hat{n}_{m\sigma}\hat{n}_{m'\sigma} \\
&= (U' - J) \sum_{m<m',\sigma} [\langle n_{m\sigma}\rangle + \delta n_{m\sigma}][\langle n_{m'\sigma}\rangle + \delta n_{m'\sigma}] \\
&= (U' - J) \sum_{m<m',\sigma} \langle n_{m\sigma}\rangle\langle n_{m'\sigma}\rangle + \langle n_{m\sigma}\rangle\delta n_{m'\sigma} + \langle n_{m'\sigma}\rangle\delta n_{m\sigma}
\end{aligned}
\tag{50}
$$

with $\delta n_{m\sigma} = \hat{n}_{m\sigma} - \langle n_{m\sigma}\rangle$

$$
\begin{aligned}
H &= (U' - J) \sum_{m<m',\sigma} \langle n_{m\sigma}\rangle\langle n_{m'\sigma}\rangle + \langle n_{m\sigma}\rangle[\hat{n}_{m'\sigma} - \langle n_{m'\sigma}\rangle] + \langle n_{m'\sigma}\rangle[\hat{n}_{m\sigma} - \langle n_{m\sigma}\rangle] \\
&= (U' - J) \sum_{m<m',\sigma} \langle n_{m\sigma}\rangle\hat{n}_{m'\sigma} + \langle n_{m'\sigma}\rangle\hat{n}_{m\sigma} - \langle n_{m\sigma}\rangle\langle n_{m'\sigma}\rangle
\end{aligned}
\tag{51}
$$

5.3 Exchange/Pair-Hopping Term

The terms are

$$
H = -J \sum_{m\neq m'} d^\dagger_{m\uparrow} d_{m\downarrow} d^\dagger_{m'\downarrow} d_{m'\uparrow} + J \sum_{m\neq m'} d^\dagger_{m\uparrow} d^\dagger_{m\downarrow} d_{m'\downarrow} d_{m'\uparrow} \tag{52}
$$

We can normal order and get

$$
H = J \sum_{m\neq m'} \left[d^\dagger_{m\uparrow} d^\dagger_{m'\downarrow} d_{m\downarrow} d_{m'\uparrow} + d^\dagger_{m\uparrow} d^\dagger_{m\downarrow} d_{m'\downarrow} d_{m'\uparrow} \right] \tag{53}
$$

Here we expand with mean field

$$
\begin{aligned}
d^\dagger_1 d^\dagger_2 d_3 d_4 = -\langle d^\dagger_1 d_3\rangle d^\dagger_2 d_4 + \langle d^\dagger_1 d_4\rangle d^\dagger_2 d_3 - \langle d^\dagger_2 d_4\rangle d^\dagger_1 d_3 + \langle d^\dagger_2 d_3\rangle d^\dagger_1 d_4 \\
+DC
\end{aligned}
\tag{54}
$$

Where DC corrects for double counting and is

$$
DC = \langle d^\dagger_1 d_4\rangle\langle d^\dagger_2 d_3\rangle - \langle d^\dagger_1 d_3\rangle\langle d^\dagger_2 d_4\rangle \tag{55}
$$

Where we drop the $\langle dd\rangle$ and the $\langle d^\dagger d^\dagger\rangle$ channels. Hence we can write

$$
H_{exc} = J \sum_{m\neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow}\rangle d^\dagger_{m'\downarrow} d_{m\downarrow} + \langle d^\dagger_{m'\downarrow} d_{m\downarrow}\rangle d^\dagger_{m\uparrow} d_{m'\uparrow} - \langle d^\dagger_{m\uparrow} d_{m\downarrow}\rangle d^\dagger_{m'\downarrow} d_{m'\uparrow} - \langle d^\dagger_{m'\downarrow} d_{m'\uparrow}\rangle d^\dagger_{m\uparrow} d_{m\downarrow} \right] \tag{56}
$$

with a correction to the energy shift given by the following

$$
\Delta E_{exc} = J \sum_{m\neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow}\rangle\langle d^\dagger_{m'\downarrow} d_{m\downarrow}\rangle - \langle d^\dagger_{m\uparrow} d_{m\downarrow}\rangle\langle d^\dagger_{m'\downarrow} d_{m'\uparrow}\rangle \right] \tag{57}
$$

Likewise we can write

$$
H_{ph} = J \sum_{m\neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow}\rangle d^\dagger_{m\downarrow} d_{m'\downarrow} + \langle d^\dagger_{m\downarrow} d_{m'\downarrow}\rangle d^\dagger_{m\uparrow} d_{m'\uparrow} - \langle d^\dagger_{m\uparrow} d_{m'\downarrow}\rangle d^\dagger_{m\downarrow} d_{m'\uparrow} - \langle d^\dagger_{m\downarrow} d_{m'\uparrow}\rangle d^\dagger_{m\uparrow} d_{m'\downarrow} \right] \tag{58}
$$

$$
\Delta E_{ph} = J \sum_{m\neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow}\rangle\langle d^\dagger_{m\downarrow} d_{m'\downarrow}\rangle - \langle d^\dagger_{m\uparrow} d_{m'\downarrow}\rangle\langle d^\dagger_{m\downarrow} d_{m'\uparrow}\rangle \right] \tag{59}
$$

6 Complete Kanamori Hamiltonian

$$
\begin{aligned}
H^k_{MF} = U \sum_m &\langle n_{m\uparrow}\rangle \hat{n}_{m\downarrow} + \langle n_{m\downarrow}\rangle \hat{n}_{m\uparrow} \\
&+ U' \sum_{m\neq m'} \langle n_{m\uparrow}\rangle \hat{n}_{m'\downarrow} + \langle n_{m'\downarrow}\rangle \hat{n}_{m\uparrow} \\
&+ (U' - J) \sum_{m<m',\sigma} \langle n_{m\sigma}\rangle \hat{n}_{m'\sigma} + \langle n_{m'\sigma}\rangle \hat{n}_{m\sigma} \\
&+ J \sum_{m\neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow}\rangle d^\dagger_{m'\downarrow} d_{m\downarrow} + \langle d^\dagger_{m'\downarrow} d_{m\downarrow}\rangle d^\dagger_{m\uparrow} d_{m'\uparrow} - \langle d^\dagger_{m\uparrow} d_{m\downarrow}\rangle d^\dagger_{m'\downarrow} d_{m'\uparrow} - \langle d^\dagger_{m'\downarrow} d_{m'\uparrow}\rangle d^\dagger_{m\uparrow} d_{m\downarrow} \right] \\
&+ J \sum_{m\neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow}\rangle d^\dagger_{m\downarrow} d_{m'\downarrow} + \langle d^\dagger_{m\downarrow} d_{m'\downarrow}\rangle d^\dagger_{m\uparrow} d_{m'\uparrow} - \langle d^\dagger_{m\uparrow} d_{m'\downarrow}\rangle d^\dagger_{m\downarrow} d_{m'\uparrow} - \langle d^\dagger_{m\downarrow} d_{m'\uparrow}\rangle d^\dagger_{m\uparrow} d_{m'\downarrow} \right]
\end{aligned} \tag{60}
$$

Where we shift the energy by

$$
\begin{aligned}
\Delta E = -U \sum_m &\langle n_{m\uparrow}\rangle \langle n_{m\downarrow}\rangle - U' \sum_{m\neq m'} \langle n_{m\uparrow}\rangle \langle n_{m'\downarrow}\rangle \\
&- (U' - J) \sum_{m<m',\sigma} \langle n_{m\sigma}\rangle \langle n_{m'\sigma}\rangle \\
&+ J \sum_{m\neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow}\rangle \langle d^\dagger_{m'\downarrow} d_{m\downarrow}\rangle - \langle d^\dagger_{m\uparrow} d_{m\downarrow}\rangle \langle d^\dagger_{m'\downarrow} d_{m'\uparrow}\rangle \right] \\
&+ J \sum_{m\neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow}\rangle \langle d^\dagger_{m\downarrow} d_{m'\downarrow}\rangle - \langle d^\dagger_{m\uparrow} d_{m'\downarrow}\rangle \langle d^\dagger_{m\downarrow} d_{m'\uparrow}\rangle \right]
\end{aligned} \tag{61}
$$

7 Basis States Ordering

Index	**Layer**	**Spin**	**Orbital**
0	L_1	$\uparrow$	d_{yz}
1	L_1	$\uparrow$	d_{xz}
2	L_1	$\uparrow$	d_{xy}
3	L_1	$\downarrow$	d_{yz}
4	L_1	$\downarrow$	d_{xz}
5	L_1	$\downarrow$	d_{xy}
6	L_2	$\uparrow$	d_{yz}
7	L_2	$\uparrow$	d_{xz}
8	L_2	$\uparrow$	d_{xy}
9	L_2	$\downarrow$	d_{yz}
10	L_2	$\downarrow$	d_{xz}
11	L_2	$\downarrow$	d_{xy}

Table 1: Basis state ordering: layer-major → spin-major → orbital-minor.